Python OOP – Classes, Objects, Attributes & Methods

Objective

In this assignment, students will practice the fundamental concepts of **Object-Oriented Programming (OOP)** in Python.

Students will work with:

- Classes
- Objects
- Attributes
- Methods
- Lists inside Classes
- Basic operations using Methods

Task 1 – Mobile Class

Question

Create a class named `Mobile`.

Attributes

The class should have:

- `brand`
- `price`

Methods

Create the following methods:

`call()`

Display a message that the mobile is making a call.

`details()`

Display the mobile's brand and price.

Example

```
Brand: Samsung
Price: ₹25000

Calling...
Brand: Samsung
Price: ₹25000
```

Concepts Covered

- Class
- Object
- Attributes
- Methods

Task 2 – Car Class

Question

Create a class named `Car`.

Attributes

The class should have:

- `brand`
- `color`
- `price`

Methods

Create the following methods:

`start()`

Display a message that the car has started.

`stop()`

Display a message that the car has stopped.

Example

```
Brand: Toyota
Color: Black
Price: ₹1500000
```

```
Car started.
Car stopped.
```

Concepts Covered

- Class
- Object
- Attributes
- Methods

Task 3 – Employee Class

Question

Create a class named `Employee`.

Attributes

The class should have:

- `name`
- `id`
- `salary`

Create a method:

```
display_details()
```

This method should display all the employee details.

Example

```
Name: Rahul
ID: 101
Salary: ₹45000
```

Concepts Covered

- Class
- Object
- Attributes
- Methods
- Displaying Object Data

Task 4 – Laptop Class

Question

Create a class named `Laptop`.

Attributes

The class should have:

- `brand`
- `model`
- `price`

Create a method:

```
apply_discount()
```

The method should reduce the laptop's price by **10%**.

Example

```
Brand: Dell
Model: Inspiron
Original Price: ₹60000

Discount: 10%
Final Price: ₹54000
```

Concepts Covered

- Class
- Object
- Attributes
- Methods
- Mathematical Operations using Object Data

Task 5 – Shopping Cart

Question

Create a class named `ShoppingCart`.

The shopping cart should store items inside a **list**.

Attribute

Create:

```
items
```

The `items` attribute should be an empty list initially.

Methods

Create the following methods:

`add_item()`

Add an item to the shopping cart.

`remove_item()`

Remove an item from the shopping cart.

`get_total()`

Calculate and return the total price of all items in the cart.

Example

```
Items in Cart:

Laptop    ₹50000
Mouse     ₹1000
Keyboard  ₹2000

Total: ₹53000
```

Concepts Covered

- Class
- Object
- Attributes
- Methods
- Lists inside Classes
- Adding and Removing Items
- Loops
- Calculating Total

Learning Outcomes

After completing these tasks, students should be able to:

1. Create a class using the `class` keyword.
2. Create objects from a class.
3. Define and use attributes.
4. Create methods inside a class.
5. Access object attributes using `self`.
6. Perform operations using methods.
7. Store multiple values using lists inside a class.
8. Understand how objects manage their own data and behaviour.